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Epileptic Disorders

ORIGINAL ARTICLE

Effectiveness and safety of cenobamate after conversion to monotherapy or dual therapy in real-world clinical practice

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Abstract

Objective: Patients with drug-resistant epilepsy (DRE) typically take multiple anti-seizure medications (ASMs) and are at risk of treatment-related adverse events (AEs). This study assessed the impact of cenobamate monotherapy or dual therapy in patients with drug-resistant epilepsy.

Methods: This was a multicenter (six Spanish epilepsy units), retrospective, observational study in patients with drug-resistant epilepsy who had received treatment with cenobamate as monotherapy (after conversion from polytherapy) or in dual therapy for at least three consecutive months, between April 2017 and November 2024.

Results: In total, 125 patients (mean age 46 ± 15.9 years; median epilepsy duration 16 years, range: 1–65) were included. Median follow-up was 11 months (range 3–80). Patients had received a median of 5 (range 2–19) previous anti-seizure medications and were receiving a median of 2 (range 0–3) concomitant anti-seizure medications at cenobamate initiation; the most common concomitant anti-seizure medications were lacosamide (36%), brivaracetam (29.6%), and clobazam (26.4%). The median cenobamate dose at the last visit was 200 mg/day (range 100–400 mg/day). By the last visit, 31.5% of patients had achieved seizure freedom, and 75.8%, 58.1%, and 49.2% had achieved ≥50%, ≥75%, and ≥90% reductions in seizure frequency, respectively. During cenobamate treatment, 52% of patients discontinued one, 32% discontinued two, and 4% discontinued three concomitant anti-seizure medications. In total, 63/125 patients (50.4%) had adverse events during cenobamate treatment, all mild or moderate; 40.8% of patients had adverse events at the last visit. The most reported adverse events were somnolence (22.4%), dizziness (10.4%), fatigue (7.2%), and instability (4.8%).

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Significance: This study highlights the real-world effectiveness and safety of cenobamate, either as monotherapy (after conversion from polytherapy) or in dual therapy, in patients with drug-resistant epilepsy. Cenobamate allows for a significant treatment simplification, resulting in fewer treatment-related adverse events without compromising seizure reduction.

KEYWORDS
antiepileptic, antiseizure, drug-resistant epilepsy, focal epilepsy

1 | INTRODUCTION

Epilepsy is a chronic neurological disease characterized by unprovoked seizures.[1] Patients with persistent seizures have an increased risk of injuries, significant comorbidities, a reduced quality of life, and are at risk of sudden unexpected death in epilepsy (SUDEP).[2–5] Seizure freedom is therefore the key target of epilepsy treatment. To this end, a plethora of antiseizure medications (ASMs) are available with different mechanisms of action, effectiveness, and tolerability profiles.[6] However, over 30% of patients with epilepsy continue to experience seizures despite treatment with at least two previous ASMs because of drug-resistant epilepsy (DRE).[7]

Recently, new ASMs have been approved that show encouraging efficacy and tolerability in refractory patients.[8–10] Cenobamate is a unique ASM with a dual mechanism of action and a well-defined pharmacology profile, inhibiting excitatory neurotransmission by blocking voltage-gated sodium channels and promoting inhibitory pathways by acting as a positive allosteric modulator of $GABA_A$ receptors.[11]

Randomized, placebo-controlled clinical trials[8,12] and open-label studies[13,14] have demonstrated the efficacy and tolerability of cenobamate in patients with uncontrolled focal seizures. These studies have been backed up by real-world evidence supporting the effectiveness and safety of cenobamate in clinical settings.[15–17] In 2021, cenobamate was approved by the European Medicines Agency for the adjunctive treatment of focal-onset seizures, with or without secondary generalization, in adult patients with epilepsy who have not achieved adequate seizure control despite treatment with at least two ASMs.[18]

Once efficacy with adjunctive cenobamate is established, most patients may be able to reduce their dose of, or even discontinue, concomitant ASMs.[19] This is significant as patients with epilepsy typically receive more than five prescription medications to manage their condition or associated comorbidities, with the number of concomitant treatments rising with age and disease severity.[20] Polypharmacy presents a considerable challenge for these patients, and especially those with uncontrolled seizures; it increases patients' risk of treatment-related AEs, significantly affects their quality of life, and leads to higher healthcare costs.[20,21] Considering the impact of polypharmacy on patients with epilepsy, the primary treatment goal should be to achieve seizure freedom with the fewest number of ASMs possible.[21] Reducing unnecessary drug load and minimizing AEs can improve treatment adherence, lower costs, and enhance quality of life. Indeed, discontinuation of ASMs in seizure-free patients has been associated with improvements in mood, cognitive function, and overall psychosocial well-being.[20]

Key points

- Cenobamate allowed to use a lower number of antiseizure medications with good tolerance and safety in a large cohort of patients with drug-resistant epilepsy.
- Simultaneously, cenobamate led to a high rate of seizure freedom (31.5%) as monotherapy or part of a dual therapy in these patients.
- The responder rate was 75.8% in this series. This gives further information about its high effectiveness beyond seizure freedom.

The aim of the present study was to assess the impact of cenobamate on seizure frequency and concomitant ASMs management in patients with DRE[22] in real-world clinical practice.

2 | METHODS

2.1 | Study design

This was a multicenter, retrospective, observational study conducted in the epilepsy units of six Spanish tertiary hospitals (Hospital Regional Universitario de Málaga, Málaga; Hospital Universitario Reina Sofía, Córdoba; Hospital Universitario San Cecilio, Granada; Hospital Universitario Lucus Augusti, Lugo; Hospital Clínico Universitario de

Valladolid, Valladolid; Hospital Universitario Virgen de la Victoria, Málaga) between April 2017 and November 2024. The study protocol was approved by the ethics committee at Hospital Regional Universitario de Málaga and was in accordance with the Declaration of Helsinki. Since this study is retrospective and data are anonymized, the need to collect informed consent from each patient was waived.

2.2 | Inclusion criteria

Patients with focal DRE were included if they had received treatment with cenobamate in monotherapy (after conversion from polytherapy) or dual therapy for at least three consecutive months (they all had tried—previously to start cenobamate—at least two ASMs without total control of their seizures).

2.3 | Data collection

The following information was collected from patients' charts: patient demographics and disease characteristics at baseline, including age, duration of epilepsy, and prior/current ASMs usage; cenobamate dosage; changes in seizure frequency between baseline (considering the 3 months prior to cenobamate initiation) and the last visit, including the proportion of patients with a ≥50%, ≥75%, ≥90%, or 100% reduction, no change, or worsening in seizure frequency; concomitant ASMs usage, including the number and type of concomitant ASMs at initiation of cenobamate and at the last visit; and AEs. Investigators from each epilepsy unit had full access to their patients' records.

2.4 | Statistical analysis

Patient and disease characteristics at baseline, cenobamate dosage, seizure frequency, concomitant ASMs usage, and AEs were reported using descriptive statistics. Continuous variables were expressed as means (±standard deviation, SD) or medians (range) according to their distribution. Categorical variables were expressed as frequencies and percentages. *StataBE 19* was used as statistical software.

3 | RESULTS

In total, out of 716 patients who were treated with cenobamate in the six participating centres between April 2017 and November 2024, 125 patients were included in the study (17.5%). After initiation of cenobamate monotherapy/dual therapy, patients were followed up for a median of 11 months, range 3–80. It is remarkable that 6 patients (4.8%) had taken part in cenobamate clinical trials and 6 (4.8%) had received cenobamate before its approval in Spain as part of a cenobamate expanded access program.

TABLE 1 Patient and disease characteristics at baseline.

Characteristic	All patients ($N = 125$)
Age, years	
Mean (SD)	46.0 (15.9)
Median (range)	47 (15–80)
Gender, *n* (%)	
Female	69 (55.2)
Male	56 (44.8)
Median duration of epilepsy (range), years[a]	16 (1–65)
Baseline seizure frequency (estimated per month and patient)	36
Number of ASMs discontinued prior to cenobamate start, *n* (%)	
0	4 (3.2)
1	22 (17.6)
2	31 (24.8)
3	15 (12)
4	16 (12.8)
5–10	33 (26.4)
11–18	4 (3.2)
Median (range)	3 (0–18)
Number of concomitant ASMs at start of cenobamate treatment, *n* (%)	
0	1 (.8)
1	31 (24.8)
2	62 (49.6)
3	30 (24)
Median (range)	2 (0–3)

Abbreviations: ASMs, antiseizure medications; SD, standard deviation.

[a]$n = 122$.

3.1 | Patient and disease characteristics at baseline

At baseline, the mean age of patients was 46 ± 15.9 years and 55.2% were female. The median duration of epilepsy was 16 years, range 1–65 (Table 1). All patients had received at least 2 ASMs before starting cenobamate treatment (median 5, range 2–19); 26/125 (21%) had received at least six prior ASMs. Patients had discontinued a median of 3 ASMs (range 0–18) prior to starting cenobamate

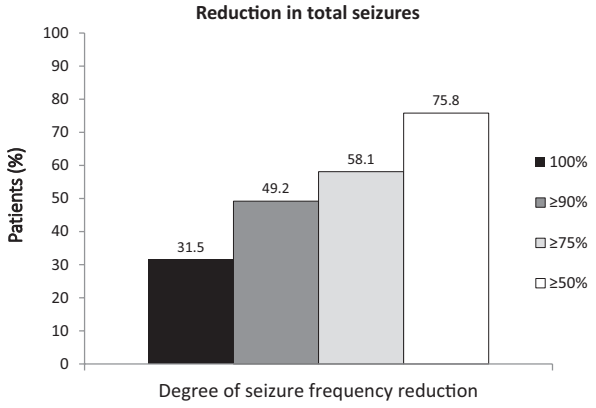


FIGURE 1 Proportion of patients with a reduction in seizure frequency (100%, ≥90%, ≥75% or ≥50%) with cenobamate between baseline and last visit.

treatment. This cohort's estimated baseline seizure frequency was 36 seizures per month and patient.

3.2 | Cenobamate dosage

Cenobamate was administered according to the label.[18] The median dose was 200 mg/day (range 100–400 mg/day).

3.3 | Effectiveness

By the last visit and compared with their seizure frequency before starting cenobamate, ≥50%, ≥75%, and ≥90% reductions in seizure frequency were achieved by 94/124 (75.8%), 72/124 (58.1%), and 61/124 (49.2%) patients, respectively; 39/124 patients (31.5%) had achieved a 100% reduction in seizure frequency (i.e., seizure freedom; Figure 1). Seizure frequency was unchanged in 15/124 patients (12.1%) and worse in 4/124 (3.2%). One patient was already seizure free when he initiated cenobamate (not included in calculations). Among patients who reached cenobamate monotherapy (28/125), one was the patient already free of seizures and he remained so, but 12/27 (44.4%) who were not seizure-free achieved seizure freedom with cenobamate.

3.4 | Concomitant ASMs usage

At the start of cenobamate treatment, patients were receiving a median of 2 (range 0–3) concomitant ASMs: 1/125 (.8%), 31/125 (24.8%), 63/125 (50.4%), and 30/125 (24%) were receiving 0, 1, 2, and 3 concomitant ASMs, respectively. Those who were receiving only one ASM had tried at least another ASM previously. The most common concomitant ASMs at the start of cenobamate treatment were lacosamide (45/125 patients; 36%), brivaracetam (37/125; 29.6%), clobazam (33/125; 26.4%), eslicarbazepine (27/125; 21.6%), and perampanel (25/125; 20%; Figure 2, Table 2).

The only patient who was taking 0 concomitant ASMs at cenobamate start had decided on her own to discontinue her previous ASMs because she thought she was getting no benefit. She suffered a seizure cluster, and then cenobamate was initiated.

During the study, 28/125 patients (22.4%) switched to cenobamate monotherapy and 97/125 (77.6%) continued under dual therapy. By the last visit, patients were receiving .78 concomitant ASMs per person, the most common of which were brivaracetam (27/125; 21.6%) and clobazam (26/125; 20.8%) (Figure 2, Table 2).

During treatment with cenobamate, the median number of discontinued ASMs was 1 (range 0–3), and 110/125 patients (88%) discontinued at least one concomitant ASM: 65/125 patients (52%) discontinued one, 40/125 (32%) discontinued two, and 5/125 (4%) discontinued three concomitant ASMs (Figure 2). Between the start of cenobamate treatment and the final visit, there were substantial reductions in the number of patients taking the sodium channel blockers (eslicarbazepine −89%, carbamazepine −89%, lamotrigine −88%, oxcarbazepine −86%, lacosamide −80%); zonisamide (80%); the AMPA receptor antagonist perampanel (80%); and the benzodiazepine clonazepam (75%); but smaller reductions (≤50%) for all other ASMs.

3.5 | Safety

Sixty-three of 125 patients (50.4%) reported AEs at any time following treatment with cenobamate. The most reported AEs were somnolence (28/125 patients; 22.4%), dizziness (13/125 patients; 10.4%), fatigue (9/125; 7.2%), and instability (6/125; 4.8%); all AEs were mild or moderate. At the last visit, 51/125 patients (40.8%) had AEs; somnolence (17/125 patients; 13.6%), dizziness (11/125 patients; 8.8%), and fatigue (11/125; 8.8%) were still the most common AEs at this timepoint (Table 3).

4 | DISCUSSION

In this multicenter, retrospective, observational study, we report the effectiveness and safety of cenobamate in a subset of patients with uncontrolled epilepsy who were able to convert from polytherapy to cenobamate monotherapy or dual therapy within real-world clinical practice. This

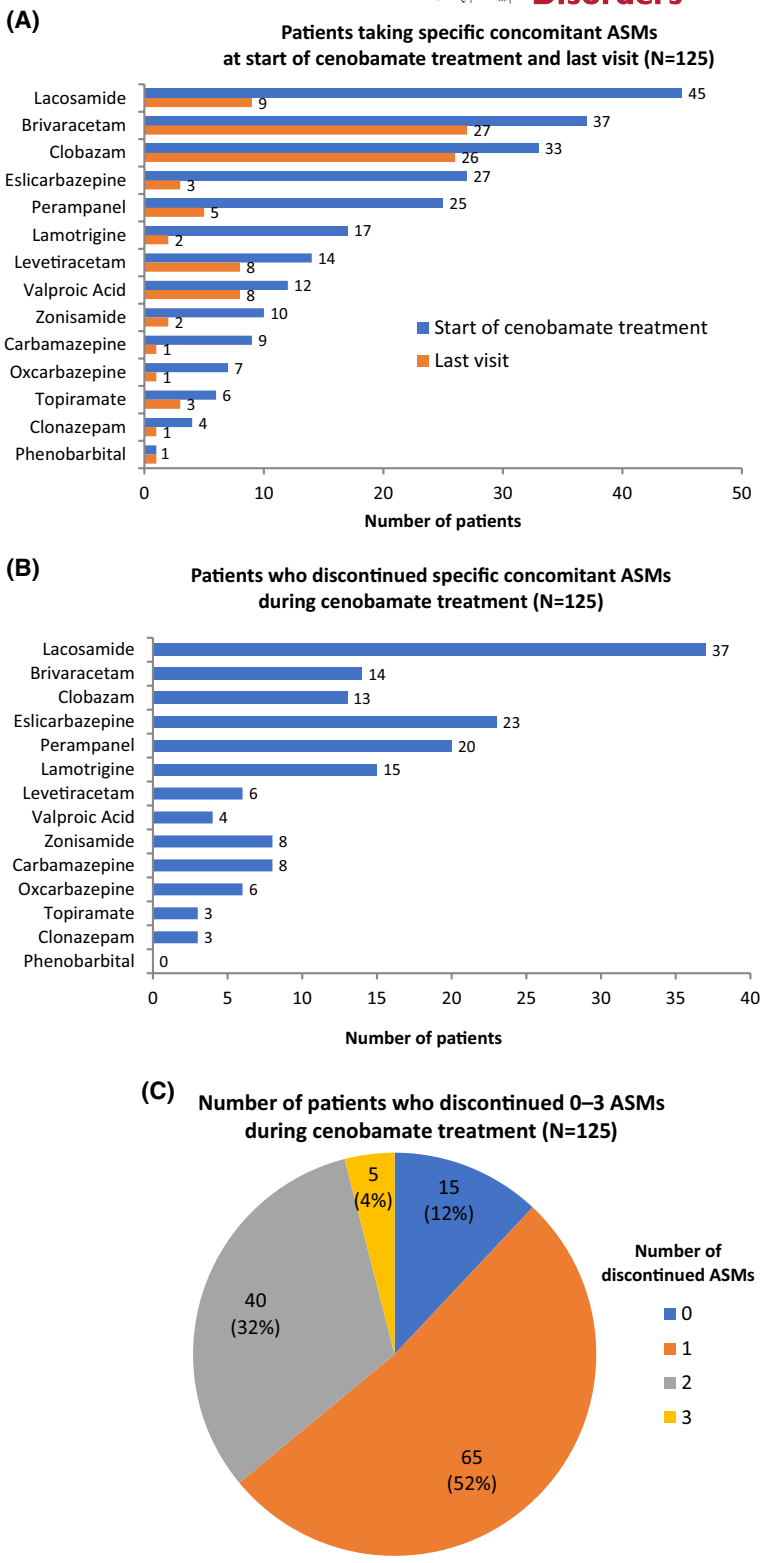


FIGURE 2 Impact of cenobamate on concomitant ASMs usage. (A) Number of patients taking specific ASMs at start of cenobamate treatment and last visit; (B) Number of patients who discontinued specific ASMs during cenobamate treatment; (C) Number of patients who discontinued 0–5 concomitant ASMs during cenobamate treatment.

adds to the growing body of evidence supporting positive outcomes with cenobamate in patients with refractory seizures, both in clinical trial[8,12,13] and real-world settings.[16,17,23–25]

In our study, we observed a high rate of seizure freedom (31.5%) with cenobamate, considerably higher than in several other real-world studies (Table 4), even though cenobamate was being used as subsequent monotherapy or dual therapy for at least three consecutive months.[17,23,26] Furthermore, almost half of patients (49%) had at least a 90% reduction in seizure frequency following cenobamate treatment, which would be expected to lead to substantial quality-of-life benefits for these patients. Our outcomes are also favorable when compared with cenobamate Phase

TABLE 2 Proportion of patients with concomitant ASMs usage during cenobamate treatment.

Patients with concomitant ASMs, *n* (%)	At start of treatment (*N* = 125)	At last follow-up (*N* = 125)	Discontinued during cenobamate treatment (*N* = 125)
Any	124 (99.2%)	97 (77.6)	109 (87.2)
Lacosamide	45 (36.0)	9 (7.2)	37 (29.6)
Brivaracetam	37 (29.6)	27 (21.6)	15 (12.0)
Clobazam	33 (26.4)	26 (20.8)	12 (9.6)
Eslicarbazepine	27 (21.6)	3 (2.4)	23 (18.4)
Perampanel	25 (20.0)	5 (4.0)	20 (16.0)
Lamotrigine	17 (13.6)	2 (1.6)	15 (12.0)
Levetiracetam	14 (11.2)	8 (6.4)	6 (4.8)
Valproic acid	12 (9.6)	8 (6.4)	4 (3.2)
Zonisamide	10 (8.0)	2 (1.6)	8 (6.4)
Carbamazepine	9 (7.2)	1 (.8)	8 (6.4)
Oxcarbazepine	7 (5.6)	1 (.8)	6 (4.8)
Topiramate	6 (4.8)	3 (2.4)	3 (2.4)
Clonazepam	4 (3.2)	1 (.8)	3 (2.4)
Phenobarbital	1 (.8)	1 (.8)	0 (0)

Abbreviation: ASMs, antiseizure medications.

2 data (28% seizure freedom at 6 weeks)[8] and similar to a post hoc analysis of a Phase III study (13% seizure freedom with a much longer 29.5 month follow-up and 36.3% seizure freedom at ≥12-months).[27]

In our study, all patients had received at least two different ASMs by the start of cenobamate treatment; 79% of patients had discontinued at least two ASMs and 21% had absolute drug resistance based on discontinuation of at least six ASMs.[28] Overall, our patients had received an average of 3.7 previous ASMs, indicating that they are a less refractory patient population compared with other cenobamate real-world cohorts (10.2–12.1 previous ASMs; Table 4[17,23,26,29]). Nonetheless, literature evidence suggests that 12%–24% of those patients who have already failed on two previous ASMs achieve seizure freedom with a third ASM, and there is a steep drop-off with subsequent ASMs.[7,30]

Thus, cenobamate may be considered as a particularly effective ASM for patients with uncontrolled focal-onset epilepsy after using two previous ASMs.[31] Furthermore, a recent study suggests that while cenobamate is more effective in patients who have received fewer prior ASMs, it can still be effective in ultra-refractory patients after more than 10 failed ASMs, providing a 42% reduction in seizure frequency and a 14% rate of seizure freedom.[32]

In a similar study to ours with other drugs, in which patients with epilepsy received only perampanel as subsequent monotherapy, 55% were reported to be seizure free for at least a 3-month period. However, this was a less refractory population than in our study; 35% of patients had not failed on any other previous ASM and a further 23% had failed on 1–2 previous ASMs.[33]

In another similar real-world study recently published in the United States, a similar responder rate (defined as the percentage of patients who achieved a seizure frequency reduction ≥50%) was found, along with a much higher seizure freedom rate (55%) for patients under cenobamate monotherapy.[34] The reason why seizure freedom rates are so different might be related to a different degree of drug resistance shown by patients in each series, since many of our patients were being treated with a higher number of ASMs when they initiated cenobamate. While that is the first real-world cohort of patients with such treatment regimen, to the best of our knowledge, ours is the first similar study in Europe focused on patients with simple therapeutic regimens (monotherapy or dual therapy) including cenobamate, and the largest one to date.

The rate of AEs in our study (50.4%) was lower than reported in cenobamate clinical trials (59%–83%)[8,12,13] and in a Spanish real-world study (74.1%).[17] Notably, the AEs rate was even lower (40.8%) at the last visit. This may reflect the lower number of concomitant ASMs in our study as patients had received cenobamate as subsequent monotherapy or dual therapy for at least three consecutive months. A low number of concomitant ASMs has many benefits for patients, including better treatment adherence, a reduced risk of treatment-related AEs, improvements in mood, cognitive function and psychosocial well-being, and lower healthcare costs.[20,21]

In patients with DRE, up to 45% reductions in the number of concomitant ASMs have been reported following treatment with cenobamate,[17,26,29] alongside reduction in the prescribed daily dose/defined daily dose ratios for

TABLE 3 Adverse events.

Patients with AEs, *n* (%)	Any timepoint (*N* = 125)	At last visit (*N* = 125)
Any	63 (50.4)	51 (40.8)
Somnolence	28 (22.4)	17 (13.6)
Dizziness	13 (10.4)	11 (8.8)
Fatigue	9 (7.2)	11 (8.8)
Instability	6 (4.8)	3 (2.4)
Attention/cognitive deficits	5 (4.0)	2 (1.6)
Headache	5 (4.0)	4 (3.2)
Irritability	4 (3.2)	2 (1.6)
Depression	4 (3.2)	4 (3.2)
Memory impairment	3 (2.4)	2 (1.6)
Bradypsychia	3 (2.4)	1 (.8)
Diplopia	2 (1.6)	1 (.8)
Possible language impairment[a]	1 (.8)	0 (.0)
Subjective gait impairment	1 (.8)	0 (.0)
Increased GGT (hemochromatosis)	1 (.8)	1 (.8)
Increased triglycerides (hemochromatosis)	1 (.8)	1 (.8)
Self-aggressiveness	1 (.8)	0 (.0)
Restless legs	1 (.8)	0 (.0)
Nausea	1 (.8)	0 (.0)
Weight loss	1 (.8)	1 (.8)
Dysarthria	1 (.8)	1 (.8)
Subjective visuospatial impairment	1 (.8)	1 (.8)

Abbreviations: AEs, adverse events; GGT, γ-glutamyl transferase.

[a]Occurred when brivaracetam dose increased.

selected ASMs.[26,29] Furthermore, a 5.6-year follow-up of cenobamate clinical trials suggests that this reduction in ASMs usage is maintained over the long term.[35] Despite the low number of concomitant ASMs used in our study, we still saw substantial reductions in ASMs usage over time. The number of patients taking SCBs, zonisamide and clonazepam fell by 75%–92% between cenobamate initiation and the final visit. This aligns with cenobamate's dual mechanism of action as a sodium channel blocker and GABA receptor modulator.[11]

However, we also observed reduction in the number of patients taking all other ASMs included in the study (except phenobarbital), including perampanel (80% reduction) and levetiracetam (43% reduction). These findings are consistent with other studies showing a reduction in concomitant ASMs usage during cenobamate treatment.[17,26,29,36] While cenobamate would be expected to reduce the need for ASMs with overlapping mechanisms of action, its antiseizure effects may also lessen the need for unrelated ASMs to sustain seizure control. Proactive dose reductions for other ASMs are now recommended during cenobamate treatment in order to optimize efficacy and tolerability.[37,38] Specifically, proactive down-titration or withdrawal is recommended when concomitant clobazam, phenytoin, phenobarbital, and SCB are taken at high dosages with cenobamate, or when a patient is receiving two or more SCB.[38]

In practice, most patients with focal onset seizures may be able to have concomitant ASMs doses lowered or discontinued once efficacy with adjunctive cenobamate is established.[19] Importantly, cenobamate efficacy is not impaired by decreasing concomitant ASMs doses, whereas tolerability may be improved. Such dose reductions also increase cenobamate retention and may therefore positively support patient quality of life and adherence.[29,35]

5 | LIMITATIONS

This study has several limitations based on its retrospective design, including the potential for selection bias or incomplete data due to unknown gaps in medical records. Information was gathered from previous medical records in six different centers, so possibly there is an unidentified lack of information in some cases. Information is not recorded

TABLE 4 Real-world outcomes with cenobamate in patients with drug-resistant epilepsy.

Characteristic	Current study (*N* = 125)	Villanueva[17] (*N* = 170)	Serrano-Castro[26] (*N* = 20)	Rodríguez-Uranga[29] (*N* = 34)	Lauxmann[23] (*N* = 116)
Mean age, years	46	39.9	40.9	36.7	38.5
Mean epilepsy duration, years	19.8	27.8	NR	23.1	NR
Mean number of previous ASMs	3.7	12.1	10.2	12.1	10.5 (median)
Mean number of coASMs at baseline	1.98	3.2	3	2.9	3 (median)
Mean number of coASMs after cenobamate treatment	.78[a]	2.6[b]; 2.2[c]	2.5[b]	1.6[c]	NR
Effectiveness (% reduction in seizure frequency) % patients					
100%	31.5	16.9[b]; 10[c]	5[a]	44.8[b]; 33.3[c]	18.4[a]
≥90%	49.2	29.1[b]; 38.5[c]	15[a]	58.6[b]; 47.6[c]	NR
≥50%	75.8	65[b]; 74.4[c]	70[a]	82.8[b]; 85.7[c]	50[a]
AEs % patients	50.4	74.1[b]	NR	52.3[c]	NR

Abbreviations: AEs, adverse events; coASMs, concomitant antiseizure medications; NR, not reported.

[a]Patients had received cenobamate as subsequent monotherapy (after conversion from polytherapy) or dual therapy for at least three consecutive months.

[b]6 months.

[c]12 months.

with the same methods or software in all the participating centers, as they belong to different regions of Spain.

Furthermore, the study has no control group, so the outcomes may not be transferable to other patient populations or healthcare settings. To compensate for this, we compare our results with other published series of patients with cenobamate, and the study assesses a relatively large patient population across six medical centers and is one of the most extensive real-world clinical assessments of cenobamate to date. Controlled, prospective studies are therefore required to confirm these findings.

Nevertheless, controlled, prospective studies are required to confirm our findings and what circumstances make it possible to simplify some patients' treatments to monotherapy or dual therapy while other patients' epilepsies remain drug-resistant.

6 | CONCLUSION

This study highlights the real-world effectiveness and safety of cenobamate as subsequent monotherapy or dual therapy in patients who have used multiple ASMs. Our data suggest that seizure freedom is achievable for many patients receiving cenobamate, and even those with few concomitant ASMs. By minimizing the need for multiple ASMs, patients receiving cenobamate can experience a low medication burden and few treatment-related AEs, without compromising seizure freedom.

AUTHOR CONTRIBUTIONS

Á. Sánchez-Guijo Benavente: study design, data collection, manuscript writing and supervision. G. García Martín: study design, data collection, patient recruitment, and manuscript supervision. E. D. Castro Ruiz: data collection and manuscript supervision. P. Cabezudo García: patient recruitment. Y. López Moreno: patient recruitment. J. C. Estévez María: patient recruitment. I. Lorite Fuentes: data collection. S. Navarro Conti: data collection. R. Calle Calle: data collection and patient recruitment. A. Fernández Cabrera: data collection and patient recruitment. I. Ros González: data collection. D. Campos Blanco: patient recruitment. M. I. Chamorro Muñoz: patient recruitment. P. Serrano Castro: study design, patient recruitment and manuscript supervision.

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CONFLICT OF INTEREST STATEMENT

Á. Sánchez-Guijo Benavente, G. García Martín, P. Cabezudo García, Y. López Moreno, J. C. Estévez María, R. Calle Calle, A. Fernández Cabrera, D. Campos Blanco, M. I. Chamorro Muñoz, and P. Serrano Castro have received research funding, consultant fees, and/or speaker honoraria from Angelini Pharma (all of them), Bial, Eisai, Esteve, GlaxoSmithKline, Jazz Pharmaceuticals, Kern Pharma, LivaNova, Nutricia, and/or UCB Pharma. The remaining authors have no conflicts of interest.

DATA AVAILABILITY STATEMENT

Anonymized data will be made available by request from qualified investigators, with approval from our ethics committee.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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Test yourself

1. Cenobamate is approved for the next types of epilepsy:
a. Focal and generalized epilepsies in adults.
b. Only generalized epilepsies in adults
c. Only focal epilepsies in adults.
d. Only generalized epilepsies in children.
2. Cenobamate, in this cohort of 125 patients, was associated most frequently with one of the next adverse effects:
a. Somnolence.
b. Dizziness.
c. Fatigue.
d. Headache.
3. Choose the right option:
a. Patients who have tried no previous antiseizure medications are the most usual candidates for cenobamate.
b. Severe adverse effects (that require hospitalization) are common in the first months of cenobamate treatment.
c. Cenobamate has often been used over long periods of time along with sodium channel blockers and benzodiazepines, so this combination is recommended for most patients.
d. Cenobamate allows for a treatment simplification in a significant percentage of patients.

Answers may be found in the Supporting information.